THE OBSTRUCTION CLASS FOR LIFTING AN ALGEBRA ALONG A SURJECTION AND THE PARAMETRIZATION OF SOLUTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the situation and the equation-to-solve fixed earlier in the chapter; the proof opens by recalling which identifications it uses and then treats the three parts.

Editorial note. The eight references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The identification of the naive cotangent complex with the truncation of the cotangent complex. Lemma lemma-relation-with-naive-cotangent-complex, used at proof lines 3185-3187. Transfers the deformation-theory statements to the cotangent complex.
- The identification H^0(L_{B/A}) = Omega_{B/A}. Lemma lemma-identify-H0 (583-586), used at proof lines 3186-3188. Makes the deformation-theory identifications apply.
- The obstruction class and parametrization of solutions in deformation theory. Deformation Theory, Lemma defos-lemma-parametrize-solutions, Remark defos-remark-parametrize-solutions, Lemmas defos-lemma-huge-diagram and defos-lemma-choices, used at proof lines 3188-3199 and 3206-3210. Supply parts (2) and (3) and the class xi' for part (1).
- The distinguished triangle for A' -> A -> B and its long exact Ext sequence. equation-triangle, used at proof lines 3216-3223. Produces the map whose image is the obstruction.

Source reference. Original lemma 08S7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11120–11120. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08S5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11118–11118. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0GPX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20842–20842. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 0GPY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20843–20843. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original equation 08SN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11136–11136. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original equation 08QS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11070–11070. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11058–11058. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08RB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11089–11089. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *In the situation above we have*

(1) *There is a canonical element* $\xi \in \mathrm{Ext}^2_B(L_{B/A}, N)$ *whose vanishing is a sufficient and necessary condition for the existence of a solution to (08SN).*
(2) *If there exists a solution, then the set of isomorphism classes of solutions is principal homogeneous under* $\mathrm{Ext}^1_B(L_{B/A}, N)$.
(3) *Given a solution* B'*, the set of automorphisms of* B' *fitting into (08SN) is canonically isomorphic to* $\mathrm{Ext}^0_B(L_{B/A}, N)$.

Proof. Via the identifications $NL_{B/A} = \tau_{\geq -1} L_{B/A}$ (Lemma 08RB) and $H^0(L_{B/A}) = \Omega_{B/A}$ (Lemma 08QF) we have seen parts (2) and (3) in Deformation Theory, Lemmas 08S5 and 08S7.

Proof of (1). Roughly speaking, this follows from the discussion in Deformation Theory, Remark 0GPY by replacing the naive cotangent complex by the full cotangent complex. Here is a more detailed explanation. By Deformation Theory, Lemma 0GPX and Remark 0GPY there exists an element

$$\xi' \in \mathrm{Ext}^1_A(NL_{A/A'}, N) = \mathrm{Ext}^1_B(NL_{A/A'} \otimes^{\mathbf{L}}_A B, N) = \mathrm{Ext}^1_B(L_{A/A'} \otimes^{\mathbf{L}}_A B, N)$$

(for the equalities see Deformation Theory, Remark 0GPY and use that $NL_{A'/A} = \tau_{\geq -1} L_{A'/A}$) such that a solution exists if and only if this element is in the image of the map

$$\mathrm{Ext}^1_B(NL_{B/A'}, N) = \mathrm{Ext}^1_B(L_{B/A'}, N) \longrightarrow \mathrm{Ext}^1_B(L_{A/A'} \otimes^{\mathbf{L}}_A B, N)$$

The distinguished triangle (08QS) for $A' \to A \to B$ gives rise to a long exact sequence

$$\ldots \to \mathrm{Ext}^1_B(L_{B/A'}, N) \to \mathrm{Ext}^1_B(L_{A/A'} \otimes^{\mathbf{L}}_A B, N) \to \mathrm{Ext}^2_B(L_{B/A}, N) \to \ldots$$

Hence taking ξ the image of ξ' works. □